

# HDCA

## Honours Diploma in Computer Application (Tally)



450 Hrs Rs.49,500/-

- Students must do two projects using Tally Prime and VBA Excel.



### OBJECTIVE

Upon completion of this course, the participant acquires the advantage in choosing his / her career as an Application Programmer, Computerised Accountant, Web Programmer.



### TARGET AUDIENCE

School final students / Graduates / Students pursuing a career in Computer.



### PREREQUISITES

No prior Computer knowledge is required.



### OVERVIEW

This course has been tailor made as per the software industry requirement. It is a stepping stone for those who aspire to be a software professional.

## COURSE COVERAGE

### I. END-USER APPLICATION

#### WINDOWS

Computer Fundamentals - Introduction to OS/GUI- Wordpad - Notepad - Paintbrush - Windows Explorer - Control Panel.

#### MS OFFICE

**Word** Creating Text and Documents - Indexes and Tables - Embedding Picture - Table Manipulation - Mail Merge - Macro.

**Excel** Formatting Cells - Charts - Pivot Table - Data Table - Forms - Scenarios - Goal seek - Macros and Module.

**PowerPoint** Creating Presentations - OLE - Enhancing Presentation - Adding Audio & Video - Special Effects.

#### INTERNET

Creating E-mail ID - Receiving & Sending E-mail - Chatting - Locating Information - Downloading.

### II. FRONT-END LANGUAGE & LIBRARY

#### HTML5

HTML Basics - New Elements - Canvas - Multimedia - Form Elements, Semantic.

#### CSS3

Syntax - Id, Class, Tag Styling - Using Text - Fonts - Links - Lists and Tables - Box Model - Text Effects - Transition - Animations.

### III. DATABASE MANAGEMENT SYSTEM

#### MySQL

Client / Server Concepts - Relational Model - Data Definition Using SQL - Modify Tables - Using ALTER TABLE Queries, Add a Column, Add an Index, Add a Primary Key - INSERT and DELETE Queries - UPDATE Queries - Using Select Queries - SELECT, WHERE, LIMIT, ORDER BY, GROUP BY, JOIN - Constraints and Indexes - Views - Expressions - Grouping - Using Functions - Math Functions, Compare String values, Work with Dates - JOIN and Subqueries - Advanced MySQL Query Processing.



## IV. FINANCIAL ACCOUNTING

### INTRODUCTION TO TallyPrime

Principles of Accounting – 5 Steps - Maintaining Company Data - F11 Features, F12 Configure.

### MASTERS

Group-Ledger-Chart of accounts-Stock Group-Stock Category-Unit of measurement-Godown-Stock Item

### TALLY VOUCHERS

Accounting, Inventory Vouchers

### REPORTS

Balance Sheet - P & L A/c - Ratio Analysis - Cash Flow, Fund flow Statements - Ageing Analysis - Movement Analysis.

### ADVANCED ACCOUNTING

Billwise Details - Cost Centres and Cost Categories - Voucher Classes and Cost centre Classes - Multiple Currencies - Bank reconciliation - Interest Calculation - Budgets & Scenario - Job Costing.

### ADVANCED INVENTORY

Order Processing - Reorder Levels - Tracking Numbers - Batch wise Details - Additional cost Details - Bill of Materials - Price List.

### SYSTEM ADMINISTRATION AND OTHER UTILITIES

Consolidation of Accounts - Tally vault - Security Control - Tally Audit - Back up and Restore - Split company Data - Export and Import of Data - ODBC Compliance.

### POINT OF SALES

Predetermination of Invoice - Multi payment Modes - Small & Smart Bills - POS Sales Register.

### MULTI LANGUAGE

Supporting OS - Translation or Transliteration - System & Phonetic Keyboard - Sourcing from other Software.

### GOODS & SERVICE TAX (GST)

Accounting of Purchase or Sales returns - Inter State Trade Accounting - Accounting of Exempted Items - Accounting Purchases from Unregistered Dealers - Accounting Input Credit on Capital Goods - GST Computation & Forms.

### TDS & TCS

Entry - Calculator - Reports - e Returns.

## PAYROLL

Introduction to Payroll - Configuring Tally for Payroll - Creation of Payroll Masters - Recording of attendance - Preparation of Payroll Vouchers based on attendance - Preparation of a Salary Disbursement Voucher - Payroll Reports (Pay slip, Pay Sheet, Payroll Statement, Payment Advice, Payroll Register, Attendance Sheet, Attendance Register) - Introduction to Provident Fund.

### ADD - ON FEATURES

WhatsApp Connect - Import from Excel - Dashboard.

### PROJECT\*

Live simulated projects are given to students.

## V. ADVANCED EXCEL & VBA PROGRAMMING

### ADVANCED EXCEL

Introduction to MS Excel and Understanding Basic Working with it - Formatting Worksheet - Working with Formulas/Functions (Lookup and Reference Functions, Logical Function, Database Functions, Date and Time Functions, Information Functions, Math and Trigonometry Functions, Statistical Functions, Text Functions) - Conditional Formatting - Connect to other Database - Data Sorting and Filtering - Pivot Tables& Charts.

### VBA PROGRAMMING

VBA Overview - VBA Programming Variables - Conditional Logic - String and String Functions - Programming Loops - Programming Arrays - Subs and Functions - Creating your first Macro - Excel VBA - Excel Objects - Excel Events - Excel VBA Error Handling - Excel VBA and User forms - Connect with an external data - Work on automation project.

### PROJECT\*

Students are given projects using VBA Excel and it will be helpful to explore how to use VBA to automate different tasks or developing applications/ tools using VBA.

## VI. DIGITAL MARKETING

Introduction to Digital Marketing, Traditional vs Digital Marketing, Organic vs Inorganic Marketing, Content Creation, Graphic Designing, Social Media Marketing, Facebook, Instagram, Youtube, Search Marketing, Google Ads, Google Analytics.

## VII. CHATGPT

ChatGPT is a free-to-use AI system. Use it for engaging conversations, gain insights, automate tasks, and witness the future of AI, all in one place.



TallyPrime



Advanced  
Excel

# HDCA

## Honours Diploma in Computer Application (Programming)



450 Hrs Rs.49,500/-

- Students must do two projects using C/C++ and Python.



### TARGET AUDIENCE

School final students / Graduates /  
Students pursuing a career in Computer.



### PREREQUISITES

No prior Computer knowledge is required.



### OVERVIEW

This course has been tailor made as per the software industry requirement. It is a stepping stone for those who aspire to be a software professional.

### OBJECTIVE



Upon completion of this course, the participant acquires the advantage in choosing his / her career as an Application Programmer, Create applications using object oriented programming concepts such as inheritance, polymorphism and virtual functions.

### COURSE COVERAGE

#### I. END-USER APPLICATION

##### WINDOWS

Computer Fundamentals - Introduction to OS/GUI- Wordpad - Notepad - Paintbrush - Windows Explorer - Control Panel.

##### MS OFFICE

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##### INTERNET

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#### II. FRONT-END LANGUAGE & LIBRARY

##### HTML5

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##### CSS3

Syntax - Id, Class, Tag Styling - Using Text - Fonts - Links - Lists and Tables - Box Model - Text Effects - Transition - Animations.

#### III. DATABASE MANAGEMENT SYSTEM

##### MySQL

Client / Server Concepts - Relational Model - Data Definition Using SQL - Modify Tables - Using ALTER TABLE Queries, Add a Column, Add an Index, Add a Primary Key - INSERT and DELETE Queries - UPDATE Queries - Using Select

Queries - SELECT, WHERE, LIMIT, ORDER BY, GROUP BY, JOIN - Constraints and Indexes - Views - Expressions - Grouping - Using Functions - Math Functions, Compare String values, Work with Dates - JOIN and Subqueries - Advanced MySQL Query Processing.

#### IV. Programming Tools

##### "C" LANGUAGE

Data Types - Operators - Expressions - Conditional Compilation - Looping & Branching - Arrays - Functions - Storage Classes - Structures and Unions - Pointers - Dynamic Memory Allocation - File Handling - Linked Lists - Double Linked List - Circular Linked List - Bit Manipulation - Introduction to Data Structure.

##### OOPS Using C++

Introduction to OOPS - Input & Output Streams - Classes & Objects - Member Functions - Function Overloading - Friend Functions & Classes - Operator Overloading - Constructor & Destructor - Inheritance - Polymorphism - File handling using Stream Classes - Instance - Templates.

##### PROJECT\*

Development of System / Graphical Utility.

##### PYTHON

Introduction to Python - History - Features - Working with Python - Basic Syntax - Variable and Data Types - Operator and Expressions - Conditional Statements - If - If- else - Nested if-else - Looping - For - While - Nested loops - Control Statements - Break - Continue - Pass - String Manipulation - Accessing Strings - Basic Operations - String slices - Function and Methods - Lists - Introduction - Accessing list - Operations - Working with lists - Function and Methods - Tuple - Introduction - Accessing tuples - Operations - Working - Functions and Methods - Dictionaries - Introduction - Accessing values in dictionaries - Working with dictionaries - Properties - Functions - Defining a function - Calling a function - Types of functions - Function Arguments - Anonymous functions - Global and local variables - Modules - Importing module - Math module - Random module - Packages - Composition - Input-Output - Printing on screen - Reading data from keyboard - Opening and closing file - Reading and writing files - Functions - Exception Handling - Exception - Except clause - OOPs concept - Class and object - Attributes - Inheritance - Overloading - Overriding - Data hiding - Polymorphism - Data Structure.



## ADVANCED PYTHON PROGRAMMING

- MySQL Database Access
- GUI Programming (Tkinter)
- Image Processing in Python with Pillow & OPENCV
- Web Scraping

### PROJECT\*

Using Python students will develop web development, data science, desktop applications, and more.

## V. DIGITAL MARKETING

Introduction to Digital Marketing, Traditional vs Digital Marketing, Organic vs Inorganic Marketing, Content Creation, Graphic Designing, Social Media Marketing, Facebook, Instagram, Youtube, Search Marketing, Google Ads, Google Analytics.

## VII. CHATGPT

ChatGPT is a free-to-use AI system. Use it for engaging conversations, gain insights, automate tasks, and witness the future of AI, all in one place.

# HDFD

## Honours Diploma in Full Stack Developer



450 Hrs Rs.64,800/-



### OBJECTIVE

After completing the program, you will be eligible for a variety of career opportunities and build your career in front-end developer, back-end developer, API, or full stack developer roles in organizations ranging from start-ups to Fortune 500 companies.



### TARGET AUDIENCE

School Final / Diploma / Graduates / Engineering



### PREREQUISITES

No prior Computer knowledge is required.



### OVERVIEW

This course is designed for beginners to specialize in program solving skills with Programming Languages and building the complete modern web sites with Front-End, Back-End and Web Framework technologies, Database Management Systems. Upon completion of this course, you are 100% fit for the campus placements.

## COURSE COVERAGE

### I. PROGRAMMING LANGUAGE

#### "C" LANGUAGE

Data Types - Operators - Expressions - Conditional Compilation - Looping & Branching - Arrays - Functions - Storage Classes - Structures and Unions - Pointers - Dynamic Memory Allocation - File Handling - Linked Lists - Double Linked List - Circular Linked List - Bit Manipulation - Introduction to Data Structure.

#### OOPS USING C++

Introduction to OOPS - Input & Output Streams - Classes & Objects - Member Functions - Function Overloading - Friend Functions & Classes - Operator Overloading - Constructor & Destructor - Inheritance - Polymorphism - File handling using Stream Classes - Instance - Templates.

#### JAVA

Data Types - Arrays - Functions - Classes - Methods - Inheritance - Polymorphism - Interfaces - Packages - Garbage Collection - Exception Handling - Data Structures - Collection - Generics - File Streams - Serialization - Multi Threading - Networking - Socket Programming - JDBC - JFC / Swing

#### PYTHON

Introduction - Conditional Statements - Looping - Control Statements - String Manipulation - Lists - Tuple - Dictionaries - Functions - Modules - Input/Output - Exception Handling - Files - Packages - OOPS Concept - Class and objects - Inheritance - Polymorphism - Data Structure - MySQL Database Access - GUI Programming (Tkinter) - Image Processing in Python with Pillow & OPENCV.

### II. FRONT-END LANGUAGE & LIBRARY

#### HTML5

HTML Basics - New Elements - Canvas - Multimedia - Form Elements, Semantic.

#### CSS3

Syntax - Id, Class, Tag Styling - Using Text - Fonts - Links - Lists and Tables - Box Model - Text Effects - Transition - Animations.

#### BOOTSTRAP

Responsive layouts with Grids - Themes - Components - Building Forms.

#### ADVANCED JAVASCRIPT

Basics, Arrays - Functions and Objects - DOM Model - Event Handling - Arrow Functions - Closures - Array Filter and Array Reduce - Regular Expression - OOPs - Promises - Async/Wait.

#### REACT JS

Creating an app - Using Templates - Components - Working with Props and State - Event Handling - Routing - Hooks - useState - useEffect - useRef - useCallback - useContext - Server Integration with API calls.

### III. DATABASE MANAGEMENT SYSTEM

#### MySQL

Client / Server Concepts - Relational Model - Data Definition Using SQL - Modify Tables - Using ALTER TABLE Queries, Add a Column, Add an Index, Add a Primary Key - INSERT and DELETE Queries - UPDATE Queries - Using Select



Queries - SELECT, WHERE, LIMIT, ORDER BY, GROUP BY, JOIN - Constraints and Indexes - Views - Expressions - Grouping - Using Functions - Math Functions, Compare String values, Work with Dates - JOIN and Subqueries - Advanced MySQL Query Processing.

## MongoDB

Introduction to NOSQL databases, CRUD operations, Data Indexing & Aggregation, Working with Unstructured Data.

## IV. WEB FRAMEWORK

### ELECTIVE - I

## DJANGO

What Is a Web Framework - The MVT Design Pattern - Django template tags - If/else - If/elif/else – For – Comments – Filters - Using templates to display data in the site – Forms Creation - CRUD operations - The Form class - Processing the form - Widgets – Validation - Forms in templates - Using

Models and ORM - Building Web APIs using Django REST - Generic Views in Django Rest – Serializers - Working with Images - Authentication with tokens - Postman - Related models - Contenttypes app.

### ELECTIVE - II

## SPRING BOOT

Spring Boot Annotations – CRUD Operations – Spring Data JPA – CrudRepository & JpaRepository – findBy methods in JPA – Custom Queries in JPA – Forms Development – Hibernate – JAVA REST API – REST API Media Types – Exception Handling – Connecting Spring Boot and React.

## PROJECT

To enhance the skills of the Student as full-stack developer they are given web development projects like E-Commerce Website, Food Delivery App, Content Management System, Project Management Dashboard, etc.